

Unit 7. Generalised Linear Models I: Categorical Responses

1 Binary Responses

FREES §11.1–11.2.

Throughout, $y_1, \dots, y_n$ are independent with $y_i \in \{0, 1\}$, $\mathbf{x}_i \in \mathbb{R}^p$ is a non-random covariate vector whose first coordinate is 1, and

$$\pi_i = \mathbf{P}\{y_i = 1 \mid \mathbf{x}_i\}, \quad \eta_i = \mathbf{x}_i' \boldsymbol{\beta}. \quad (1)$$

Since y_i is Bernoulli, $\mathbf{E}[y_i] = \pi_i$ and $\text{var}(y_i) = \pi_i(1 - \pi_i)$: the mean determines the whole distribution, and the variance is a function of the mean rather than a free parameter.

1.1 The linear probability model

Definition 7.1 (Linear probability model). The **linear probability model** takes $\pi_i = \mathbf{x}_i' \boldsymbol{\beta}$ and estimates $\boldsymbol{\beta}$ by least squares.

Proposition 7.2 (Three defects). (i) If $\mathbf{x}' \boldsymbol{\beta} \in [0, 1]$ for every $\mathbf{x} \in \mathbb{R}^p$ then $\beta_1 = \dots = \beta_{p-1} = 0$, the intercept alone surviving. On a design of finite extent the fitted values need not lie in $[0, 1]$ either.

- (ii) The errors are heteroscedastic by construction: $\text{var}(y_i - \mathbf{x}'_i \boldsymbol{\beta}) = \mathbf{x}'_i \boldsymbol{\beta} (1 - \mathbf{x}'_i \boldsymbol{\beta})$, which varies with i unless every π_i is the same.
- (iii) If a fitted value falls outside $[0, 1]$ the variance the model implies for that observation is negative.

Proof. (i) Suppose $\mathbf{x}'\boldsymbol{\beta} \in [0, 1]$ for all $\mathbf{x}$ and let $j \geq 1$ with $\beta_j \neq 0$. Taking $\mathbf{x} = (1, 0, \dots, t, \dots, 0)'$ with t in position j gives $\beta_0 + t\beta_j \in [0, 1]$ for every $t \in \mathbb{R}$, which is false since the left side is unbounded in t .

(ii) y_i is Bernoulli with mean $\pi_i = \mathbf{x}'_i \boldsymbol{\beta}$, so its variance is $\pi_i(1 - \pi_i)$, and the error $y_i - \mathbf{x}'_i \boldsymbol{\beta}$ has the same variance.

(iii) $t(1 - t) < 0$ exactly when $t < 0$ or $t > 1$. □

Remark. Only (ii) is repairable inside the linear model: the weighted least squares and robust standard errors of Unit 4 apply, with weights $1/\{\pi_i(1 - \pi_i)\}$. Defects (i) and (iii) say the model is not a probability model at all. What replaces it is a model in which the mean is a fixed increasing transform of $\mathbf{x}'_i \boldsymbol{\beta}$ with range $(0, 1)$.

1.2 Logistic and probit models

Definition 7.3 (Binary regression). Let $F : \mathbb{R} \rightarrow (0, 1)$ be a continuous, strictly increasing distribution function. The **binary regression model with response function F** is

$$\pi_i = F(\mathbf{x}_i' \boldsymbol{\beta}), \quad \text{equivalently} \quad F^{-1}(\pi_i) = \mathbf{x}_i' \boldsymbol{\beta}, \quad (2)$$

and $g = F^{-1}$ is the **link function**. The two cases used here are

$$F(z) = \Lambda(z) = \frac{1}{1 + e^{-z}} \quad (\textbf{logistic}), \quad F(z) = \Phi(z) \quad (\textbf{probit}). \quad (3)$$

Definition 7.4 (Odds and logit). For $p \in (0, 1)$ the **odds** are $p/(1 - p)$ and the **logit** is $\text{logit}(p) = \ln\{p/(1 - p)\}$.

Proposition 7.5 (The logit is the inverse of Λ). Λ is a strictly increasing distribution function on $\mathbb{R}$ with $\Lambda(-z) = 1 - \Lambda(z)$, $\Lambda' = \Lambda(1 - \Lambda)$, and $\text{logit}(\Lambda(z)) = z$. Hence the logistic model is $\text{logit}(\pi_i) = \mathbf{x}_i' \boldsymbol{\beta}$.

Proof. $\Lambda'(z) = e^{-z}/(1 + e^{-z})^2 > 0$, and $\Lambda(z) \rightarrow 0, 1$ as $z \rightarrow \mp\infty$, so Λ is a strictly increasing distribution function. Multiplying numerator and denominator of $\Lambda(-z) = 1/(1 + e^z)$ by e^{-z} gives $e^{-z}/(1 + e^{-z}) = 1 - \Lambda(z)$. Also $\Lambda(1 - \Lambda) = \{1/(1 + e^{-z})\}\{e^{-z}/(1 + e^{-z})\} = \Lambda'$. Finally $\Lambda/(1 - \Lambda) = e^z$, whose logarithm is z . $\square$

Theorem 7.6 (Threshold representation). Let $y_i^* = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ with ε_i independent, each with distribution function F symmetric about 0, and let $y_i = \mathbb{1}\{y_i^* > 0\}$. Then $\mathbf{P}\{y_i = 1\} = F(\mathbf{x}_i' \boldsymbol{\beta})$, which is (2).

Proof. $\mathbf{P}\{y_i = 1\} = \mathbf{P}\{\varepsilon_i > -\mathbf{x}_i' \boldsymbol{\beta}\} = 1 - F(-\mathbf{x}_i' \boldsymbol{\beta}) = F(\mathbf{x}_i' \boldsymbol{\beta})$, the last step by symmetry, $F(-a) = 1 - F(a)$, which holds for Φ and, by Proposition 7.5, for Λ . $\square$

Proposition 7.7 (Random utility). Let $\varepsilon_1, \varepsilon_2$ be independent with the standard Gumbel distribution function $G(a) = \exp(-e^{-a})$. Then $\varepsilon_2 - \varepsilon_1$ has the standard logistic distribution. Consequently, if an individual with utilities $U_j = V_j + \varepsilon_j$ chooses $j = 1$ when $U_1 > U_2$, then $\mathbf{P}\{\text{choose } 1\} = \Lambda(V_1 - V_2)$.

Proof. Conditioning on ε_1 and writing $\gamma(a) = G'(a) = e^{-a} \exp(-e^{-a})$,

$$\mathbf{P}\{\varepsilon_2 - \varepsilon_1 \leq t\} = \int_{-\infty}^{\infty} G(t+a) \gamma(a) da = \int_{-\infty}^{\infty} \exp(-e^{-t}e^{-a}) e^{-a} \exp(-e^{-a}) da.$$

Substitute $u = e^{-a}$, so $du = -e^{-a}da$ and u runs over $(0, \infty)$:

$$\int_0^{\infty} \exp(-u(1 + e^{-t})) du = \frac{1}{1 + e^{-t}} = \Lambda(t).$$

For the consequence, $\mathbf{P}\{U_1 > U_2\} = \mathbf{P}\{\varepsilon_2 - \varepsilon_1 < V_1 - V_2\}$. □

Theorem 7.8 (Odds ratio interpretation). In the logistic model:

(i) if $x_{ij} \in \{0, 1\}$ and the other coordinates are held fixed, then

$$e^{\beta_j} = \frac{\text{odds}\{y_i = 1 \mid x_{ij} = 1\}}{\text{odds}\{y_i = 1 \mid x_{ij} = 0\}}, \quad (4)$$

the **odds ratio** for that covariate;

- (ii) if x_{ij} is continuous, $\beta_j = \partial \ln \text{odds}\{y_i = 1\} / \partial x_{ij}$, so a unit increase multiplies the odds by e^{β_j} ;
- (iii) the effect on the **probability** is $\partial \pi_i / \partial x_{ij} = \beta_j \pi_i (1 - \pi_i)$, which depends on $\mathbf{x}_i$ and is largest at $\pi_i = 1/2$.

Proof. By Proposition 7.5 the model says $\ln \text{odds}\{y_i = 1 \mid \mathbf{x}_i\} = \mathbf{x}_i' \boldsymbol{\beta}$. For (i), subtract that identity at $x_{ij} = 0$ from the same identity at $x_{ij} = 1$, the other coordinates being equal: the difference of the left sides is the logarithm of the ratio in (4) and the difference of the right sides is β_j . For (ii), differentiate the identity in x_{ij} . For (iii), $\pi_i = \Lambda(\mathbf{x}_i' \boldsymbol{\beta})$ and $\Lambda' = \Lambda(1 - \Lambda)$ by Proposition 7.5, so the chain rule gives $\partial \pi_i / \partial x_{ij} = \beta_j \Lambda'(\eta_i) = \beta_j \pi_i (1 - \pi_i)$, and $t(1 - t)$ is maximised at $t = 1/2$. $\square$

Remark. A coefficient in a logistic regression is constant on the log-odds scale and not on the probability scale, which is the price of forcing the fitted probability into $(0, 1)$. The same covariate change moves the probability by $\beta_j \pi(1 - \pi)$, four times as much at $\pi = 1/2$ as at $\pi = 0.1$.

Proposition 7.9 (Logit against probit). Λ and Φ are both symmetric about 0 with value $1/2$ there.

- (i) Their slopes at 0 agree when the logistic coefficient is $4/\sqrt{2\pi} = 1.5958$ times the probit one.
- (ii) The standard logistic distribution has variance $\pi^2/3$, so matching the standard deviation of the latent error in Theorem 7.6 makes the logistic coefficient $\pi/\sqrt{3} = 1.8138$ times the probit one.

Proof. (i) $\Lambda'(0) = \Lambda(0)\{1 - \Lambda(0)\} = 1/4$ and $\Phi'(0) = 1/\sqrt{2\pi}$. The models $\Lambda(\beta_L z)$ and $\Phi(\beta_P z)$ have derivatives $\beta_L/4$ and $\beta_P/\sqrt{2\pi}$ at $z = 0$, equal when $\beta_L = 4\beta_P/\sqrt{2\pi}$.

(ii) The standard logistic density is $\Lambda'(z) = e^{-z}/(1 + e^{-z})^2$, symmetric, so the mean is 0 and the variance is $\int z^2 \Lambda'(z) dz$. Substituting $u = \Lambda(z)$, so $z = \ln\{u/(1 - u)\}$ and $du = \Lambda' dz$,

$$\int_{\mathbb{R}} z^2 \Lambda'(z) dz = \int_0^1 \left(\ln \frac{u}{1-u} \right)^2 du = 2 \int_0^1 (\ln u)^2 du - 2 \int_0^1 (\ln u) \{\ln(1-u)\} du,$$

after expanding the square and using the symmetry $u \mapsto 1 - u$ to identify the two squared terms. The first integral is 2 by two integrations by parts. For the second, expand $\ln(1 - u) =$

$-\sum_{m \geq 1} u^m/m$ and integrate term by term, justified by monotone convergence since every term is of one sign:

$$-\int_0^1 (\ln u) \ln(1-u) \, du = \sum_{m \geq 1} \frac{1}{m} \int_0^1 u^m \ln u \, du = -\sum_{m \geq 1} \frac{1}{m(m+1)^2}.$$

Since $\frac{1}{m(m+1)^2} = \frac{1}{m} - \frac{1}{m+1} - \frac{1}{(m+1)^2}$ and the two pieces sum to $1 - (\pi^2/6 - 1) = 2 - \pi^2/6$, the displayed integral is $\pi^2/6 - 2$ and the second term of the variance, $-2 \int_0^1 (\ln u) \ln(1-u) \, du$, is $2(\pi^2/6 - 2)$. The variance is therefore $2 \cdot 2 + 2(\pi^2/6 - 2) = \pi^2/3$. Matching sd against the probit's 1 rescales the coefficient by $\pi/\sqrt{3}$. $\square$

Remark. The two factors, 1.60 and 1.81, calibrate different things: the first the slope at $\pi = 1/2$, the second the scale of the latent error. Neither is exact, since Λ and Φ are genuinely different functions; §4.3 reports the ratio actually observed on a data set whose events are rare.

1.3 The likelihood

Definition 7.10 (Log-likelihood). For the model (2),

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left[y_i \ln F(\mathbf{x}'_i \boldsymbol{\beta}) + (1 - y_i) \ln \{1 - F(\mathbf{x}'_i \boldsymbol{\beta})\} \right]. \quad (5)$$

Theorem 7.11 (Score and information). Write $\eta_i = \mathbf{x}'_i \boldsymbol{\beta}$, $\pi_i = F(\eta_i)$ and $f = F'$. Then

$$\nabla \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{x}_i \frac{f(\eta_i)}{\pi_i(1 - \pi_i)} (y_i - \pi_i), \quad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{f(\eta_i)^2}{\pi_i(1 - \pi_i)} \mathbf{x}_i \mathbf{x}'_i, \quad (6)$$

and in the logistic case these reduce to

$$\nabla \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{x}_i (y_i - \pi_i) = \mathbf{X}'(\mathbf{Y} - \boldsymbol{\pi}), \quad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^n \pi_i(1 - \pi_i) \mathbf{x}_i \mathbf{x}'_i = \mathbf{X}' \mathbf{W} \mathbf{X}, \quad (7)$$

with $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$. In the logistic case $-\nabla^2 \ell(\boldsymbol{\beta}) = \mathcal{I}(\boldsymbol{\beta})$ identically, not merely in expectation.

Proof. Differentiating (5) through η_i and using $\partial\eta_i/\partial\boldsymbol{\beta} = \mathbf{x}_i$,

$$\nabla\ell = \sum_i \mathbf{x}_i \left\{ \frac{y_i}{\pi_i} - \frac{1-y_i}{1-\pi_i} \right\} f(\eta_i) = \sum_i \mathbf{x}_i \frac{y_i(1-\pi_i) - (1-y_i)\pi_i}{\pi_i(1-\pi_i)} f(\eta_i),$$

and the numerator is $y_i - \pi_i$. The information is the variance of the score; the summands are independent with mean zero, and $\text{var}(y_i) = \pi_i(1-\pi_i)$, so

$$\mathcal{I}(\boldsymbol{\beta}) = \sum_i \frac{f(\eta_i)^2}{\pi_i^2(1-\pi_i)^2} \text{var}(y_i) \mathbf{x}_i \mathbf{x}_i' = \sum_i \frac{f(\eta_i)^2}{\pi_i(1-\pi_i)} \mathbf{x}_i \mathbf{x}_i'.$$

For the logistic case $f = \Lambda' = \Lambda(1-\Lambda) = \pi_i(1-\pi_i)$ by Proposition 7.5, so the ratio $f/\{\pi(1-\pi)\}$ is 1 and $f^2/\{\pi(1-\pi)\} = \pi_i(1-\pi_i)$. Differentiating the logistic score once more, $\partial\pi_i/\partial\boldsymbol{\beta} = \pi_i(1-\pi_i)\mathbf{x}_i$ gives $\nabla^2\ell = -\sum_i \pi_i(1-\pi_i)\mathbf{x}_i\mathbf{x}_i'$, which involves no y_i and therefore equals its own expectation. $\square$

Corollary 7.12 (What the score equations force). At a logistic maximum likelihood estimate $\hat{\boldsymbol{\beta}}$ with fitted probabilities $\hat{\pi}_i$, every column $\mathbf{x}_{\cdot j}$ of the design satisfies $\sum_i x_{ij}\hat{\pi}_i = \sum_i x_{ij}y_i$. With an

intercept this gives $\sum_i \hat{\pi}_i = \sum_i y_i$, and for a 0/1 covariate the fitted and observed counts agree within each of its two groups.

Proof. Setting (7) to zero is $\mathbf{X}'(\mathbf{Y} - \hat{\boldsymbol{\pi}}) = \mathbf{0}$, which read coordinate by coordinate is the stated identity. The intercept column is $\mathbf{1}$, and a 0/1 column is the indicator of a group. $\square$

Theorem 7.13 (Concavity). The log-likelihood (5) is concave in $\boldsymbol{\beta}$ for the logistic and the probit response functions, and strictly concave when $\text{rank}(\mathbf{X}) = p$.

Proof. The map $\boldsymbol{\beta} \mapsto \eta_i = \mathbf{x}'_i \boldsymbol{\beta}$ is linear, so it suffices that $\eta \mapsto \ln F(\eta)$ and $\eta \mapsto \ln\{1 - F(\eta)\}$ are concave; a concave function of a linear map is concave, and a sum of concave functions is concave.

Logistic. $\ln \Lambda(\eta) = -\ln(1+e^{-\eta})$ has second derivative $-\Lambda(\eta)\{1-\Lambda(\eta)\} < 0$, and $\ln\{1-\Lambda(\eta)\} = \ln \Lambda(-\eta)$ has the same second derivative at $-\eta$.

Probit. By symmetry it is enough to treat $\ln \Phi$. Writing $\phi = \Phi'$,

$$(\ln \Phi)''(\eta) = \frac{\phi' \Phi - \phi^2}{\Phi^2} = -\frac{\eta \phi \Phi + \phi^2}{\Phi^2},$$

using $\phi'(\eta) = -\eta\phi(\eta)$, so the claim is $\eta\Phi(\eta) + \phi(\eta) > 0$ for every η . For $\eta \geq 0$ both terms are positive. For $\eta = -t$ with $t > 0$ the claim is $\phi(t) > t\{1 - \Phi(t)\}$, and

$$1 - \Phi(t) = \int_t^\infty \phi(s) \, ds < \int_t^\infty \frac{s}{t} \phi(s) \, ds = \frac{1}{t} [-\phi(s)]_t^\infty = \frac{\phi(t)}{t},$$

the inequality because $s/t > 1$ on (t, ∞) and $\phi > 0$.

Strictness. In both cases the second derivative of the per-observation term is strictly negative, so $-\nabla^2 \ell(\boldsymbol{\beta}) = \sum_i c_i \mathbf{x}_i \mathbf{x}_i'$ with every $c_i > 0$; this is positive definite exactly when the $\mathbf{x}_i$ span $\mathbb{R}^p$, that is when $\text{rank}(\mathbf{X}) = p$. $\square$

Theorem 7.14 (Separation destroys the estimate). Suppose there is $\mathbf{c} \neq \mathbf{0}$ with

$$\mathbf{x}_i' \mathbf{c} > 0 \quad \text{whenever } y_i = 1, \quad \mathbf{x}_i' \mathbf{c} < 0 \quad \text{whenever } y_i = 0. \quad (8)$$

Then, for the logistic model, $\ell(t\mathbf{c})$ increases strictly to 0 as $t \rightarrow \infty$, so $\sup_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}) = 0$ is not attained and no maximum likelihood estimate exists.

Proof. Using $\ln \Lambda(z) = -\ln(1 + e^{-z})$ and $\ln\{1 - \Lambda(z)\} = -\ln(1 + e^z)$,

$$\ell(t\mathbf{c}) = - \sum_{i:y_i=1} \ln(1 + e^{-t\mathbf{x}'_i\mathbf{c}}) - \sum_{i:y_i=0} \ln(1 + e^{t\mathbf{x}'_i\mathbf{c}}).$$

Under (8) every exponent is $-t|\mathbf{x}'_i\mathbf{c}|$, so each term is $-\ln(1 + e^{-t|\mathbf{x}'_i\mathbf{c}|})$, strictly increasing in t and tending to 0. Hence $\ell(t\mathbf{c}) \uparrow 0$. Since $\ell < 0$ everywhere, the supremum 0 is not attained. $\square$

Remark. Separation is a property of the data, not of the model, and it becomes more likely as p grows relative to n or as a category becomes rare. Software does not fail: it reports a large coefficient with a larger standard error, a residual deviance near zero and fitted probabilities equal to 0 and 1 to machine precision (§5.1).

1.4 Iteratively reweighted least squares

Theorem 7.15 (The Newton step is a weighted least squares step). For the logistic model let $\boldsymbol{\beta}^{(m)}$ be the current value, and put $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}^{(m)}$, $\pi_i = \Lambda(\eta_i)$, $w_i = \pi_i(1 - \pi_i)$ and the **working response**

$$z_i = \eta_i + \frac{y_i - \pi_i}{w_i}. \quad (9)$$

Then the Newton–Raphson update $\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \mathcal{I}(\boldsymbol{\beta}^{(m)})^{-1} \nabla \ell(\boldsymbol{\beta}^{(m)})$ is exactly the weighted least squares fit of $\mathbf{z}$ on $\mathbf{X}$ with weights w_i :

$$\boldsymbol{\beta}^{(m+1)} = (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{z}. \quad (10)$$

Proof. By Theorem 7.11, $\nabla \ell = \mathbf{X}'(\mathbf{Y} - \boldsymbol{\pi})$ and $\mathcal{I} = \mathbf{X}' \mathbf{W} \mathbf{X}$, so

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}'(\mathbf{Y} - \boldsymbol{\pi}).$$

Also $\mathbf{W} \mathbf{z} = \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(m)} + (\mathbf{Y} - \boldsymbol{\pi})$ by (9), so the right side of (10) is $(\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \{ \mathbf{X}' \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(m)} + \mathbf{X}'(\mathbf{Y} - \boldsymbol{\pi}) \}$, which is the same vector. $\square$

Remark. Because the logistic observed and expected information coincide (Theorem 7.11), Newton–Raphson and Fisher scoring are the same algorithm here, and by Theorem 7.13 the objective is concave, so the iteration has one stationary point to find. The weights $w_i = \pi_i(1 - \pi_i)$ are largest at $\pi_i = 1/2$: the observations that carry information are those whose outcome is genuinely uncertain.

2 Likelihood Inference and Classification

FREES §11.3, §11.9.

2.1 Asymptotics for the logistic estimator

Theorem 7.16 (Sampling distribution of $\hat{\beta}$). Consider the logistic model with true parameter β_0 and suppose the design satisfies

(D1) $|\mathbf{x}_i| \leq M$ for every i , with M free of n ;

(D2) $\mathbf{F}_n = n^{-1} \sum_{i \leq n} \pi_i(1 - \pi_i) \mathbf{x}_i \mathbf{x}_i' \rightarrow \mathbf{F}$ positive definite, the π_i being evaluated at β_0 .

Then with probability tending to one the maximum likelihood estimate $\hat{\beta}$ exists and is unique,

$\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1}), \quad n^{-1}\mathcal{I}(\hat{\boldsymbol{\beta}}) \xrightarrow{p} \mathbf{F}. \quad (11)$$

Proof. The three hypotheses of the concave maximum likelihood theorem of the prerequisites are checked in turn, with $\Theta = \mathbb{R}^p$ and $\ell_i(\boldsymbol{\beta}) = y_i \ln \Lambda(\mathbf{x}'_i \boldsymbol{\beta}) + (1 - y_i) \ln \{1 - \Lambda(\mathbf{x}'_i \boldsymbol{\beta})\}$.

Concavity and smoothness. Each ℓ_i is concave by Theorem 7.13 and infinitely differentiable, and Θ is all of $\mathbb{R}^p$. Strict concavity, hence uniqueness, holds when $\text{rank}(\mathbf{X}) = p$, which (D2) forces for large n : a rank deficiency would make $\mathbf{F}_n$ singular.

The score condition. By Theorem 7.11, $\nabla \ell_i(\boldsymbol{\beta}_0) = \mathbf{x}_i(y_i - \pi_i)$, which has mean zero and modulus at most $|\mathbf{x}_i| \leq M$ since $|y_i - \pi_i| \leq 1$. Its variance is $\pi_i(1 - \pi_i)\mathbf{x}_i\mathbf{x}'_i$, so the average variance is $\mathbf{F}_n \rightarrow \mathbf{F}$ by (D2).

The curvature condition. By Theorem 7.11 the negative Hessian $\mathbf{H}_n(\boldsymbol{\beta}) = \sum_i w(\mathbf{x}'_i \boldsymbol{\beta})\mathbf{x}_i\mathbf{x}'_i$, with $w(\eta) = \Lambda(\eta)\{1 - \Lambda(\eta)\}$, carries no y_i ; at $\boldsymbol{\beta}_0$ it equals $n\mathbf{F}_n$, so $n^{-1}\mathbf{H}_n(\boldsymbol{\beta}_0) \rightarrow \mathbf{F}$ deterministically. For the Lipschitz bound, $w' = \Lambda(1 - \Lambda)(1 - 2\Lambda)$ has $|w'| \leq \frac{1}{4}$, so by the mean value theorem

and Cauchy–Schwarz $|w(\mathbf{x}'_i\boldsymbol{\beta}) - w(\mathbf{x}'_i\boldsymbol{\beta}_0)| \leq \frac{1}{4}M|\boldsymbol{\beta} - \boldsymbol{\beta}_0|$, and since $\|\mathbf{x}_i\mathbf{x}'_i\| = |\mathbf{x}_i|^2 \leq M^2$,

$$\|\mathbf{H}_n(\boldsymbol{\beta}) - \mathbf{H}_n(\boldsymbol{\beta}_0)\| \leq \sum_{i=1}^n |w(\mathbf{x}'_i\boldsymbol{\beta}) - w(\mathbf{x}'_i\boldsymbol{\beta}_0)| \|\mathbf{x}_i\mathbf{x}'_i\| \leq n \frac{1}{4}M^3 |\boldsymbol{\beta} - \boldsymbol{\beta}_0|.$$

So the curvature condition holds with Lipschitz constant $M^3/4$, and the limiting curvature is the same $\mathbf{F}$ as the limiting score variance: the information equality holds here because the model is correctly specified. The conclusions are the existence, consistency, asymptotic normality and information statements of that theorem. $\square$

Remark. In practice $\mathbf{F}$ is replaced by $n^{-1}\mathcal{I}(\hat{\boldsymbol{\beta}})$, which is legitimate by the last statement of (11) and Slutsky’s theorem, and one reports $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}_0, \mathcal{I}(\hat{\boldsymbol{\beta}})^{-1})$, with $\text{se}(\hat{\beta}_j)$ the square root of the j -th diagonal entry of $\mathcal{I}(\hat{\boldsymbol{\beta}})^{-1}$. There is no exact t distribution here and no residual degrees of freedom correction: the reference distribution is normal, and the printed ratio $\hat{\beta}_j/\text{se}(\hat{\beta}_j)$ is a z statistic.

2.2 Wald, likelihood ratio, and deviance

Definition 7.17 (Test statistics). Let $\mathbf{R}$ be $q \times p$ of rank q and consider $H_0 : \mathbf{R}\boldsymbol{\beta}_0 = \mathbf{r}$. Write $\hat{\mathbf{V}} = \mathcal{I}(\hat{\boldsymbol{\beta}})^{-1}$ for the estimated variance of $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}^R$ for the maximiser of ℓ under the restriction. The **Wald** and **likelihood ratio** statistics are

$$W = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'(\mathbf{R}\hat{\mathbf{V}}\mathbf{R}')^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}), \quad LR = 2\{\ell(\hat{\boldsymbol{\beta}}) - \ell(\hat{\boldsymbol{\beta}}^R)\}. \quad (12)$$

Corollary 7.18 (Both are χ_q^2). Under the conditions of Theorem 7.16 and H_0 , both W and LR converge in distribution to χ_q^2 . For $q = 1$ and $H_0 : \beta_j = 0$ the Wald statistic is the square of the reported z statistic.

Proof. Theorem 7.16 supplies the hypotheses of the Wald corollary and of the likelihood ratio theorem of the prerequisites, the latter after the change of coordinates that turns $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ into the vanishing of q coordinates. For $q = 1$, $W = (\hat{\beta}_j - 0)^2/\hat{V}_{jj} = \{\hat{\beta}_j/\text{se}(\hat{\beta}_j)\}^2$. $\square$

Definition 7.19 (Deviance). The **deviance** of a fitted binary regression is $D = 2\{\ell_{\text{sat}} - \ell(\hat{\boldsymbol{\beta}})\}$, where ℓ_{sat} is the largest log-likelihood attainable by any set of fitted probabilities. The **null deviance** is the deviance of the intercept-only fit.

Proposition 7.20 (Deviance for ungrouped binary data). If each $y_i \in \{0, 1\}$ then $\ell_{\text{sat}} = 0$ and $D = -2\ell(\hat{\boldsymbol{\beta}})$. Consequently the difference of the deviances of two nested models equals their likelihood ratio statistic, while D itself is a function of the fitted probabilities alone.

Proof. The saturated fit puts $\hat{\pi}_i = y_i$, giving per-observation log-likelihood $y_i \ln y_i + (1 - y_i) \ln(1 - y_i) = 0$ with the convention $0 \ln 0 = 0$; and no set of probabilities can do better, since each term of (5) is a logarithm of a number in $(0, 1]$ and hence at most 0. So $\ell_{\text{sat}} = 0$ and $D = -2\ell(\hat{\boldsymbol{\beta}})$. For nested models, $D_R - D_U = -2\ell(\hat{\boldsymbol{\beta}}^R) + 2\ell(\hat{\boldsymbol{\beta}}) = LR$. $\square$

Remark. The saturated model has one parameter per observation, so the number of parameters grows with n and the asymptotics of the prerequisites do not apply to D itself: D is not a goodness-of-fit statistic for ungrouped binary data, and comparing it with $n - p$ is meaningless (§4.2 prints

the two). **Differences** of deviances between nested models are likelihood ratio statistics, and those are χ^2 by Corollary 7.18.

Definition 7.21 (Information criteria). $\text{AIC} = -2\ell(\hat{\beta}) + 2p$ and $\text{BIC} = -2\ell(\hat{\beta}) + p \ln n$, with p the number of estimated parameters.

Remark. These are the general-likelihood forms. The Gaussian linear model of Unit 5 gave the special case $\text{AIC} = (\text{SSE} + 2d\hat{\sigma}^2)/n$ after dropping a constant and dividing by n , by the proposition there relating $-2\ell + 2d$ to C_p ; the constant and the scale differ, so AIC values are comparable across models fitted to the same response by the same likelihood, and across nothing else.

2.3 Classification

Definition 7.22 (Classifier, confusion matrix). A **score** is a function $s(\mathbf{x}) \in \mathbb{R}$; the associated **classifier at threshold** t predicts $\hat{y} = \mathbf{1}\{s(\mathbf{x}) > t\}$. Its performance on a sample is recorded in the counts

	$y = 0$	$y = 1$
$\hat{y} = 0$	TN	FN
$\hat{y} = 1$	FP	TP

and the summaries

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \quad (13)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{accuracy} = \frac{\text{TP} + \text{TN}}{n}. \quad (14)$$

The **false positive rate** is $1 - \text{specificity}$.

Theorem 7.23 (Cost-optimal threshold). Let $(\mathbf{X}, y)$ have $P\{y = 1 \mid \mathbf{X} = \mathbf{x}\} = p(\mathbf{x})$, and charge $c_{01} > 0$ for predicting 0 when $y = 1$ and $c_{10} > 0$ for predicting 1 when $y = 0$. Among all classifiers

$\hat{y} = h(\mathbf{X})$, the expected cost $c_{01} \mathbf{P}\{h(\mathbf{X}) = 0, y = 1\} + c_{10} \mathbf{P}\{h(\mathbf{X}) = 1, y = 0\}$ is minimised by

$$h^*(\mathbf{x}) = \mathbf{1}\{p(\mathbf{x}) > t^*\}, \quad t^* = \frac{c_{10}}{c_{10} + c_{01}}. \quad (15)$$

Proof. Condition on $\mathbf{X} = \mathbf{x}$. Predicting 1 costs $c_{10}\{1 - p(\mathbf{x})\}$ in expectation and predicting 0 costs $c_{01}p(\mathbf{x})$. The first is smaller exactly when $c_{10} - c_{10}p(\mathbf{x}) < c_{01}p(\mathbf{x})$, that is $p(\mathbf{x}) > c_{10}/(c_{10} + c_{01})$. Since the total expected cost is the expectation over $\mathbf{X}$ of the conditional cost, minimising it pointwise minimises it overall. $\square$

Corollary 7.24. With equal costs $t^* = 1/2$ and h^* is the Bayes classifier of Unit 1, which minimises the error rate.

Remark. The threshold $1/2$ is a statement about costs, not a default. When one error is ten times as expensive as the other, $t^* = 1/11$. Under class imbalance the two are very different: a rule with threshold $1/2$ on a response that is positive 3% of the time can have high accuracy and low sensitivity at once, since predicting 0 always is already 97% accurate.

2.4 ROC and AUC

Definition 7.25 (ROC curve, AUC). Let S^+ and S^- denote the score of a random positive and a random negative case, with distribution functions F^+ and F^- . The **receiver operating characteristic** is the curve traced by

$$(\text{FPR}(t), \text{TPR}(t)) = (1 - F^-(t), 1 - F^+(t)), \quad t \in \mathbb{R}, \quad (16)$$

as t decreases from $+\infty$ to $-\infty$, and AUC is the area under it.

Theorem 7.26 (The area is a probability). If F^- is continuous and S^+, S^- are independent, then

$$\text{AUC} = \mathbf{P}\{S^+ > S^-\}. \quad (17)$$

Proof. Parametrise the curve by $u = \text{FPR}(t) = 1 - F^-(t)$, which increases from 0 to 1 as t

decreases, continuously by hypothesis. The area under the curve is

$$\text{AUC} = \int_0^1 \text{TPR} \, du = \int_{\mathbb{R}} \mathbf{P}\{S^+ > t\} \, dF^-(t) = \mathbf{E}[g(S^-)] = \mathbf{P}\{S^+ > S^-\},$$

where $g(t) = \mathbf{P}\{S^+ > t\}$. The change of variable is $u = 1 - F^-(t)$ with $du = dF^-(t)$, the orientation reversing twice; the third equality is the definition of the expectation of $g(S^-)$; and the last is Fubini's theorem applied to the indicator of $\{S^+ > S^-\}$ under the product law of the independent pair (S^+, S^-) . $\square$

Corollary 7.27 (Invariance and calibration). (i) If h is strictly increasing then the scores s and $h \circ s$ have the same ROC curve and the same AUC.
(ii) If S^+ and S^- have the same distribution, $\text{AUC} = 1/2$.
(iii) AUC does not depend on the proportion of positives in the sample.

Proof. (i) $\{h(S^+) > h(S^-)\} = \{S^+ > S^-\}$ because h is strictly increasing, so Theorem 7.26 gives the same number; the curve itself is the same set of points, reached at the transformed

thresholds. (ii) By symmetry $\mathbf{P}\{S^+ > S^-\} = \mathbf{P}\{S^- > S^+\}$ and, F^- being continuous, the two sum to 1. (iii) Both F^+ and F^- are conditional distributions given the class, so neither involves the class proportions. $\square$

Remark. Part (i) is why AUC measures **ranking** and not calibration: replacing every fitted probability $\hat{\pi}$ by $\hat{\pi}/10$ leaves AUC untouched and destroys every probability statement the model makes. Part (iii) is why AUC is stable under imbalance while precision and accuracy are not: precision has FP in its denominator, and the number of negatives available to be misclassified grows with the imbalance.

3 Multi-category Responses

FREES §11.5–11.6.

Now y_i takes values in $\{1, \dots, c\}$ with $c \geq 2$, and $\pi_{ij} = \mathbf{P}\{y_i = j \mid \mathbf{x}_i\}$, so that $\sum_{j=1}^c \pi_{ij} = 1$. Write $y_{ij} = \mathbb{1}\{y_i = j\}$. Since the c indicators of one observation sum to 1, the likelihood of that observation

is $\prod_j \pi_{ij}^{y_{ij}}$ and

$$\ell = \sum_{i=1}^n \sum_{j=1}^c y_{ij} \ln \pi_{ij}. \quad (18)$$

3.1 Nominal responses: the generalised logit

Definition 7.28 (Generalised logit). The **generalised logit** model is

$$\pi_{ij} = \frac{\exp(\mathbf{x}'_i \boldsymbol{\beta}_j)}{\sum_{k=1}^c \exp(\mathbf{x}'_i \boldsymbol{\beta}_k)}, \quad j = 1, \dots, c. \quad (19)$$

Proposition 7.29 (Identifiability and the baseline). Replacing every $\boldsymbol{\beta}_j$ by $\boldsymbol{\beta}_j + \mathbf{a}$ leaves (19) unchanged, for any $\mathbf{a} \in \mathbb{R}^p$. Hence only the differences $\boldsymbol{\beta}_j - \boldsymbol{\beta}_c$ are identified, the normalisation $\boldsymbol{\beta}_c = \mathbf{0}$ costs nothing, and under it the model has $(c-1)p$ free parameters and

$$\ln \frac{\pi_{ij}}{\pi_{ic}} = \mathbf{x}'_i \boldsymbol{\beta}_j, \quad j = 1, \dots, c-1. \quad (20)$$

Proof. Multiplying numerator and denominator of (19) by $\exp(-\mathbf{x}'_i \mathbf{a})$ turns $\boldsymbol{\beta}_k$ into $\boldsymbol{\beta}_k + \mathbf{a}$ throughout without changing the value. Taking $\mathbf{a} = -\boldsymbol{\beta}_c$ makes the last vector zero. The ratio π_{ij}/π_{ic} is $\exp(\mathbf{x}'_i \boldsymbol{\beta}_j)/\exp(\mathbf{x}'_i \boldsymbol{\beta}_c) = \exp(\mathbf{x}'_i \boldsymbol{\beta}_j)$ under that normalisation. $\square$

Remark. So a coefficient in a generalised logit is a log odds **against the baseline category**, and its sign says nothing about $\mathbf{P}\{y = j\}$ on its own: raising $\mathbf{x}'_i \boldsymbol{\beta}_j$ raises π_{ij} only relative to π_{ic} , while every other π_{ik} falls. Which category is the baseline is a labelling choice, and the fitted probabilities do not depend on it.

Proposition 7.30 ($c = 2$ is logistic regression). With $c = 2$ and $\boldsymbol{\beta}_2 = \mathbf{0}$, (19) is $\pi_{i1} = \Lambda(\mathbf{x}'_i \boldsymbol{\beta}_1)$.

Proof. $\pi_{i1} = e^{\mathbf{x}'_i \boldsymbol{\beta}_1} / (e^{\mathbf{x}'_i \boldsymbol{\beta}_1} + 1) = 1 / (1 + e^{-\mathbf{x}'_i \boldsymbol{\beta}_1})$. $\square$

Theorem 7.31 (Score and concavity). Under the normalisation $\boldsymbol{\beta}_c = \mathbf{0}$, set

$$\boldsymbol{\beta} = (\boldsymbol{\beta}'_1, \dots, \boldsymbol{\beta}'_{c-1})'.$$

Then

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}_j} = \sum_{i=1}^n \mathbf{x}_i (y_{ij} - \pi_{ij}), \quad j = 1, \dots, c-1, \quad (21)$$

and ℓ is concave in $\boldsymbol{\beta}$.

Proof. Under the normalisation, $\ln \pi_{ij} = \mathbf{x}'_i \boldsymbol{\beta}_j - A(\mathbf{V}_i)$ with $\mathbf{V}_i = (\mathbf{x}'_i \boldsymbol{\beta}_1, \dots, \mathbf{x}'_i \boldsymbol{\beta}_{c-1})'$ and $A(\mathbf{v}) = \ln(1 + \sum_{k < c} e^{v_k})$. Since $\sum_j y_{ij} = 1$, (18) becomes

$$\ell = \sum_{i=1}^n \left\{ \sum_{j < c} y_{ij} \mathbf{x}'_i \boldsymbol{\beta}_j - A(\mathbf{V}_i) \right\}. \quad (22)$$

A direct computation gives $\partial A / \partial v_j = e^{v_j} / (1 + \sum_{k < c} e^{v_k}) = \pi_{ij}$, so the chain rule and $\partial \mathbf{V}_i / \partial \boldsymbol{\beta}_j = \mathbf{x}_i$ give (21).

For concavity, the map $\boldsymbol{\beta} \mapsto \mathbf{V}_i$ is linear, so it is enough that A is convex. Differentiating once

more, $\partial^2 A / \partial v_j \partial v_k = \pi_{ij}(\delta_{jk} - \pi_{ik})$, so for $\mathbf{u} \in \mathbb{R}^{c-1}$,

$$\mathbf{u}' \nabla^2 A \mathbf{u} = \sum_{j < c} \pi_{ij} u_j^2 - \left(\sum_{j < c} \pi_{ij} u_j \right)^2 = \text{var}(U) \geq 0,$$

where U is the random variable taking the value u_j with probability π_{ij} for $j < c$ and the value 0 with probability π_{ic} : the first sum is $\mathbf{E}[U^2]$ and the second is $(\mathbf{E}[U])^2$. So A is convex, $-A$ concave, and the linear part of (22) contributes nothing to curvature. $\square$

Corollary 7.32. At the maximum likelihood estimate, $\sum_i x_{il} \hat{\pi}_{ij} = \sum_i x_{il} y_{ij}$ for every covariate l and every $j < c$. With an intercept, the fitted category counts equal the observed ones.

Proof. Set (21) to zero and read coordinate l . With $x_{i0} \equiv 1$ this is $\sum_i \hat{\pi}_{ij} = \sum_i y_{ij}$ for $j < c$, and the count for category c follows because both sides sum to n over j . $\square$

Theorem 7.33 (Independence of irrelevant alternatives). Under (19), for any two categories

$$j \neq h, \quad \frac{\pi_{ij}}{\pi_{ih}} = \exp\{\mathbf{x}'_i(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)\}, \quad \mathbf{P}\{y_i = j \mid y_i \in \{j, h\}\} = \Lambda(\mathbf{x}'_i(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)), \quad (23)$$

neither of which involves any other category, its covariates, or its presence in the choice set.

Proof. The denominator of (19) cancels in the ratio, giving the first identity. For the second, $\mathbf{P}\{y_i = j \mid y_i \in \{j, h\}\} = \pi_{ij}/(\pi_{ij} + \pi_{ih})$; dividing numerator and denominator by π_{ih} and using the first identity gives $e^u/(1 + e^u) = \Lambda(u)$ with $u = \mathbf{x}'_i(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)$. $\square$

Remark. The second identity of (23) is a specification check that costs one extra fit: restrict the sample to the two categories, run a binary logistic regression, and compare its coefficients with $\hat{\boldsymbol{\beta}}_j - \hat{\boldsymbol{\beta}}_h$ from the full fit. Both estimate the same vector when the model holds (§4.5).

The same identity is the model's substantive weakness. If two of the categories are close substitutes, adding one of them to the choice set should draw almost entirely from the other, and (23) forbids that: the odds between any two categories are unaffected by what else is available.

3.2 Ordinal responses: the cumulative logit

Definition 7.34 (Proportional odds model). Let $\alpha_1 < \alpha_2 < \cdots < \alpha_{c-1}$ and $\boldsymbol{\beta} \in \mathbb{R}^k$, the covariate vector $\mathbf{x}_i$ now carrying no intercept. The **cumulative logit**, or **proportional odds**, model is

$$\text{logit } P\{y_i \leq j \mid \mathbf{x}_i\} = \alpha_j + \mathbf{x}_i' \boldsymbol{\beta}, \quad j = 1, \dots, c-1. \quad (24)$$

Proposition 7.35 (It is a probability model exactly when the cut points are ordered). Equation (24) determines $\pi_{ij} = \Lambda(\alpha_j + \mathbf{x}_i' \boldsymbol{\beta}) - \Lambda(\alpha_{j-1} + \mathbf{x}_i' \boldsymbol{\beta})$ with the conventions $\alpha_0 = -\infty$, $\alpha_c = +\infty$. Every π_{ij} is positive for every $\mathbf{x}_i$ if and only if $\alpha_1 < \cdots < \alpha_{c-1}$.

Proof. $\pi_{ij} = P\{y_i \leq j\} - P\{y_i \leq j-1\}$, and applying Λ to both sides of (24) gives the displayed difference. Λ is strictly increasing by Proposition 7.5, so the difference is positive exactly when $\alpha_j > \alpha_{j-1}$; and this must hold for every j and every $\mathbf{x}_i$, the value of $\mathbf{x}_i' \boldsymbol{\beta}$ being irrelevant since it is added to both cut points. $\square$

Theorem 7.36 (Latent representation). Let $y_i^* = -\mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ with ε_i standard logistic, and set $y_i = j$ when $\alpha_{j-1} < y_i^* \leq \alpha_j$. Then (24) holds. Taking ε_i standard normal instead gives the **cumulative probit** model $P\{y_i \leq j\} = \Phi(\alpha_j + \mathbf{x}_i' \boldsymbol{\beta})$.

Proof. $P\{y_i \leq j\} = P\{y_i^* \leq \alpha_j\} = P\{\varepsilon_i \leq \alpha_j + \mathbf{x}_i' \boldsymbol{\beta}\} = \Lambda(\alpha_j + \mathbf{x}_i' \boldsymbol{\beta})$, and applying logit gives (24). The probit statement is the same computation with Φ in place of Λ . $\square$

Theorem 7.37 (Why “proportional odds”). Under (24), for every cut point j the cumulative odds satisfy

$$\frac{\text{odds}\{y_i \leq j \mid x_{il} + 1\}}{\text{odds}\{y_i \leq j \mid x_{il}\}} = e^{\beta_l}, \quad (25)$$

the other covariates held fixed. The ratio is the same at every j : that constancy is the content of the model, and it is a restriction, not a consequence of the ordering.

Proof. By (24), $\ln \text{odds}\{y_i \leq j\} = \alpha_j + \mathbf{x}_i' \boldsymbol{\beta}$. Subtracting the same identity with x_{il} in place of $x_{il} + 1$ leaves β_l , and α_j cancels; exponentiating gives (25), free of j . $\square$

Corollary 7.38 (A diagnostic). Under (24), for each j the dichotomised response $\mathbf{1}\{y_i \leq j\}$ follows a binary logistic regression on $\mathbf{x}_i$ with intercept α_j and slope vector $\boldsymbol{\beta}$, the **same** $\boldsymbol{\beta}$ for every j . Fitting the $c - 1$ dichotomisations separately therefore produces $c - 1$ estimates of one vector.

Proof. $P\{\mathbf{1}\{y_i \leq j\} = 1 \mid \mathbf{x}_i\} = \Lambda(\alpha_j + \mathbf{x}_i' \boldsymbol{\beta})$ by Proposition 7.35, which is Definition 7.3 with response function Λ and linear predictor $\alpha_j + \mathbf{x}_i' \boldsymbol{\beta}$. $\square$

Remark. Corollary 7.38 is how proportional odds is checked: the $c - 1$ separate slope vectors should agree up to sampling error, and a systematic drift across j says the restriction is wrong. The generalised logit of §3.1 is not a nesting model for the proportional odds model, since neither family contains the other, so the two are compared by an information criterion rather than by a likelihood ratio test (§4.6).

Remark (Two sign conventions). FREES writes the model as in (24), with $+\mathbf{x}_i' \boldsymbol{\beta}$. R's `MASS::polr` fits logit $P\{y \leq j\} = \zeta_j - \mathbf{x}_i' \boldsymbol{\gamma}$, so that $\boldsymbol{\gamma} = -\boldsymbol{\beta}$ and $\zeta_j = \alpha_j$: a positive `polr` coefficient pushes the response towards **higher** categories. The two parametrisations give the

same fitted probabilities and the same log-likelihood, and §4.6 checks that they do.

4 Computation: Formulas Against Package Output

FREES §11.4.

Binary data: **Default** from ISLR2, default on balance, income and student status. Multi-category data: **Wage** from ISLR2, marital status (nominal) and education (ordinal).

```
1 suppressMessages({library(ISLR2); library(MASS); library(nnet)
2                       library(sandwich)})
3 d <- Default; n <- nrow(d); y <- as.numeric(d$default == "Yes")
4 ctl <- glm.control(epsilon = 1e-14, maxit = 100) # tighter than the default
5 c(n = n, defaults = sum(y), rate = mean(y))
```

```
##           n    defaults      rate
## 10000.0000   333.0000    0.0333
```

4.1 The linear probability model (§1.1)

```
1 lp <- lm(y ~ balance + income + student, data = d); fv <- fitted(lp)
2 c(min_fitted = min(fv), max_fitted = max(fv),
```

```

3 fitted_outside_01 = sum(fv < 0 | fv > 1),
4 implied_variance_negative = sum(fv*(1 - fv) < 0), of = n)

```

```

##                min_fitted                max_fitted
##                -0.08965                0.26506
## fitted_outside_01 implied_variance_negative
##                3134.00000                3134.00000
##                of
##                10000.00000

```

Nearly a third of the fitted probabilities are negative, and by Proposition 7.2(iii) each of those carries a negative implied variance. The heteroscedasticity is real but repairable, by the robust standard errors of Unit 4.

```

1 rb <- sqrt(diag(vcovHC(lp, type = "HC1")))
2 round(cbind(OLS_se = coef(summary(lp))[, 2], robust_se = rb,
3             ratio = rb/coef(summary(lp))[, 2]), 6)

```

```

##                OLS_se robust_se  ratio
## (Intercept) 0.008382  0.008635 1.0302
## balance      0.000004  0.000007 1.8722
## income       0.000000  0.000000 0.9701

```

```
## studentYes  0.005663  0.005674 1.0020
```

4.2 Score, information, IRLS (§1.3, §1.4)

```
1 g <- glm(y ~ balance + income + student, data = d, family = binomial,
2         control = ctl)
3 X <- model.matrix(g); b <- coef(g); p <- drop(plogis(X %*% b))
4 U <- drop(crossprod(X, y - p))           # score:  $X'(y - \hat{p})$ 
5 Info <- t(X) %*% (p*(1 - p) * X)         # information:  $X'WX$ 
6 c(max_abs_score = max(abs(U)), largest_column_entry = max(abs(X)),
7   se_manual_vs_glm = max(abs(sqrt(diag(solve(Info)))) -
8                           summary(g)$coefficients[, 2])),
9   smallest_se = min(summary(g)$coefficients[, 2]))
```

```
##          max_abs_score largest_column_entry
##          1.585e-08          7.355e+04
## se_manual_vs_glm          smallest_se
##          8.972e-10          8.203e-06
```

Corollary 7.12 is an exact arithmetic identity at the optimum, not an approximation: the fitted probabilities reproduce the observed counts overall and within each level of any 0/1 covariate.

```

1 c(defaults = sum(y), fitted_total = sum(p),
2   student_defaults = sum(y[d$student == "Yes"]),
3   student_fitted = sum(p[d$student == "Yes"]))

```

```

##          defaults      fitted_total student_defaults
##          333          333          127
## student_fitted
##          127

```

Theorem 7.15 says the whole fit is a sequence of weighted least squares steps.

```

1 irls <- function(X, y, tol = 1e-13) {
2    b <- rep(0, ncol(X))
3    repeat {
4      eta <- drop(X %*% b); mu <- plogis(eta); w <- mu*(1 - mu)
5      z  <- eta + (y - mu)/w                                     # working response
6      bn <- drop(solve(t(X) %*% (w*X), crossprod(X, w*z)))
7      if (max(abs(bn - b)) < tol) return(bn)
8      b <- bn
9    }
10 }

```

```
11 c(irls_vs_glm = max(abs(irls(X, y) - b)), coefficient_scale = max(abs(b)))
```

```
##           irls_vs_glm coefficient_scale
##           9.592e-14          1.087e+01
```

4.3 Odds ratios, and probit against logit (§1.2)

A unit of **balance** is one dollar, so its odds ratio per unit is uninformative; Theorem 7.8 applies to any increment, and one standard deviation is the readable one.

```
1 sdx <- c(1, sd(d$balance), sd(d$income), 1)
2 round(rbind(coefficient = b, odds_ratio_per_unit = exp(b),
3             odds_ratio_per_sd = exp(b*sdx)), 4)
```

```
##                (Intercept) balance income studentYes
## coefficient             -10.87  0.0057  0.000        -0.6468
## odds_ratio_per_unit         0.00  1.0058  1.000         0.5237
## odds_ratio_per_sd          0.00 16.0360  1.041         0.5237
```

```
1 gp <- glm(y ~ balance + income + student, data = d, control = ctl,
2           family = binomial(link = "probit"))
3 round(rbind(logit = b, probit = coef(gp), ratio = b/coef(gp)), 5)
```

```
##          (Intercept) balance income studentYes
## logit      -10.869  0.00574   0.000    -0.6468
## probit     -5.475  0.00282   0.000    -0.2960
## ratio       1.985  2.03366   1.444     2.1852
```

```
1 c(slope_match_factor = 4/sqrt(2*pi), sd_match_factor = pi/sqrt(3))
```

```
## slope_match_factor    sd_match_factor
##                1.596                1.814
```

Neither calibration factor of Proposition 7.9 reproduces the observed ratios. The three coefficients that are estimated with any precision give ratios near 2; **income** has $p = 0.71$ in both fits, so its ratio of 1.44 is noise. The two response functions differ most in the tails, and this response is positive 3.3% of the time, so the fit is determined far from $\pi = 1/2$, which is where both factors are computed.

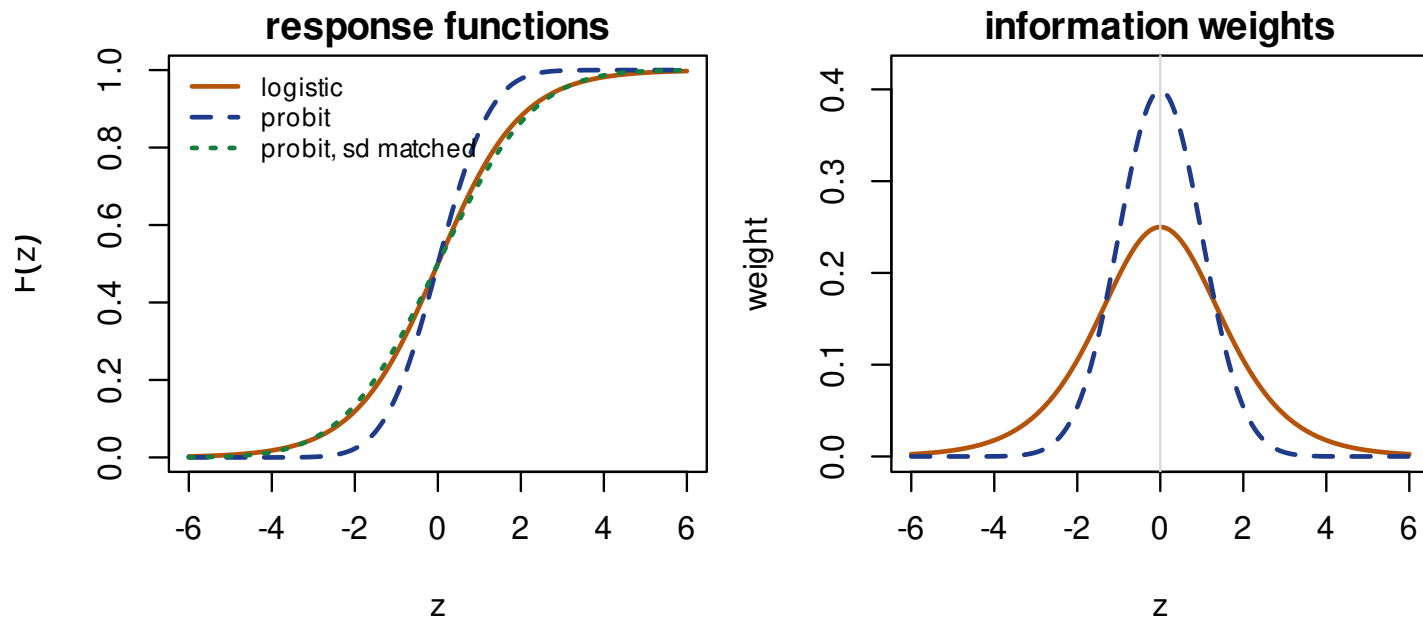


Figure 7.1: Left: the two response functions of (3), and the probit rescaled by $\pi/\sqrt{3}$ so that the latent errors have equal standard deviation (Proposition 7.9). Right: the weights w_i each observation receives in the information matrix of Theorem 7.11, largest where the outcome is most uncertain.

4.4 Wald, likelihood ratio, deviance (§2.1–§2.2)

```

1 g0 <- glm(y ~ balance + income, data = d, family = binomial, control = ctl)
2 V <- solve(Info); j <- which(names(b) == "studentYes")
3 W <- unname(b[j]^2 / V[j, j]); LR <- as.numeric(2*(logLik(g) - logLik(g0)))
4 c(Wald = W, z_statistic_squared = summary(g)$coefficients[j, 3]^2,
5   LR = LR, deviance_drop = deviance(g0) - deviance(g),
6   p_Wald = pchisq(W, 1, lower.tail = FALSE),
7   p_LR = pchisq(LR, 1, lower.tail = FALSE))

```

##	Wald	z_statistic_squared	LR
##	7.494427	7.494427	7.421443
##	deviance_drop	p_Wald	p_LR
##	7.421443	0.006189	0.006445

The two statistics test the same hypothesis and are both χ_1^2 in the limit (Corollary 7.18), but they are different numbers in finite samples, here 7.494 against 7.421. The likelihood ratio equals the drop in deviance exactly, which is Proposition 7.20.

```

1 c(deviance = deviance(g), minus_2_loglik = -2*as.numeric(logLik(g)),
2   residual_df = n - length(b), null_deviance = g$null.deviance,

```

```
3 AIC_manual = -2*as.numeric(logLik(g)) + 2*length(b), AIC = AIC(g)
```

```
##      deviance minus_2_loglik    residual_df null_deviance
##      1572          1572          9996          2921
##      AIC_manual          AIC
##      1580          1580
```

The deviance is $-2\ell(\hat{\beta})$ to machine precision, as Proposition 7.20 requires, and it is nowhere near its residual degrees of freedom: for ungrouped binary data that comparison carries no information about fit.

4.5 Classification and ROC (§2.3–§2.4)

```
1 s <- fitted(g)
2 tab <- t(sapply(c(0.5, 0.2, 1/11, mean(y)), function(t) {
3   yh <- as.numeric(s > t)
4   TN <- sum(yh == 0 & y == 0); FP <- sum(yh == 1 & y == 0)
5   FN <- sum(yh == 0 & y == 1); TP <- sum(yh == 1 & y == 1)
6   c(threshold = t, TN = TN, FP = FP, FN = FN, TP = TP,
7     sensitivity = TP/(TP + FN), specificity = TN/(TN + FP),
8     precision = TP/(TP + FP), accuracy = (TP + TN)/length(y)) })))
```

```
9 round(tab, 4)
```

```
##      threshold  TN   FP  FN  TP sensitivity specificity
## [1,]    0.5000 9627   40 228 105      0.3153      0.9959
## [2,]    0.2000 9390  277 130 203      0.6096      0.9713
## [3,]    0.0909 9059  608   80 253      0.7598      0.9371
## [4,]    0.0333 8373 1294   35 298      0.8949      0.8661
##      precision accuracy
## [1,]    0.7241    0.9732
## [2,]    0.4229    0.9593
## [3,]    0.2938    0.9312
## [4,]    0.1872    0.8671
```

```
1 c(accuracy_of_predicting_zero_always = 1 - mean(y))
```

```
## accuracy_of_predicting_zero_always
##                                0.9667
```

At the conventional threshold the rule finds 31.5% of the defaults and is 97.3% accurate, against 96.7% for the rule that never predicts a default: almost all of the accuracy comes from the class proportions. The third row is the cost-optimal threshold of Theorem 7.23 when a missed default costs ten times a

false alarm.

Theorem 7.26 and Corollary 7.27 give five routes to one number.

```

1 np <- sum(y == 1); nn <- sum(y == 0)
2 rank_auc <- function(v) (sum(rank(v)[y == 1]) - np*(np + 1)/2)/(np*nn)
3 thr <- sort(unique(c(-Inf, s, Inf)), decreasing = TRUE)
4 tpr <- sapply(thr, function(t) mean(s[y == 1] > t))
5 fpr <- sapply(thr, function(t) mean(s[y == 0] > t))
6 c(rank_form = rank_auc(s),
7   wilcoxon = unname(wilcox.test(s[y == 1], s[y == 0])$statistic)/(np*nn),
8   trapezoid = sum(diff(fpr)*(head(tpr, -1) + tail(tpr, -1))/2),
9   on_the_logit_scale = rank_auc(log(s/(1 - s))),
10  after_dividing_by_ten = rank_auc(s/10))

```

```

##           rank_form           wilcoxon
##           0.9496           0.9496
##           trapezoid   on_the_logit_scale
##           0.9496           0.9496
## after_dividing_by_ten
##           0.9496

```

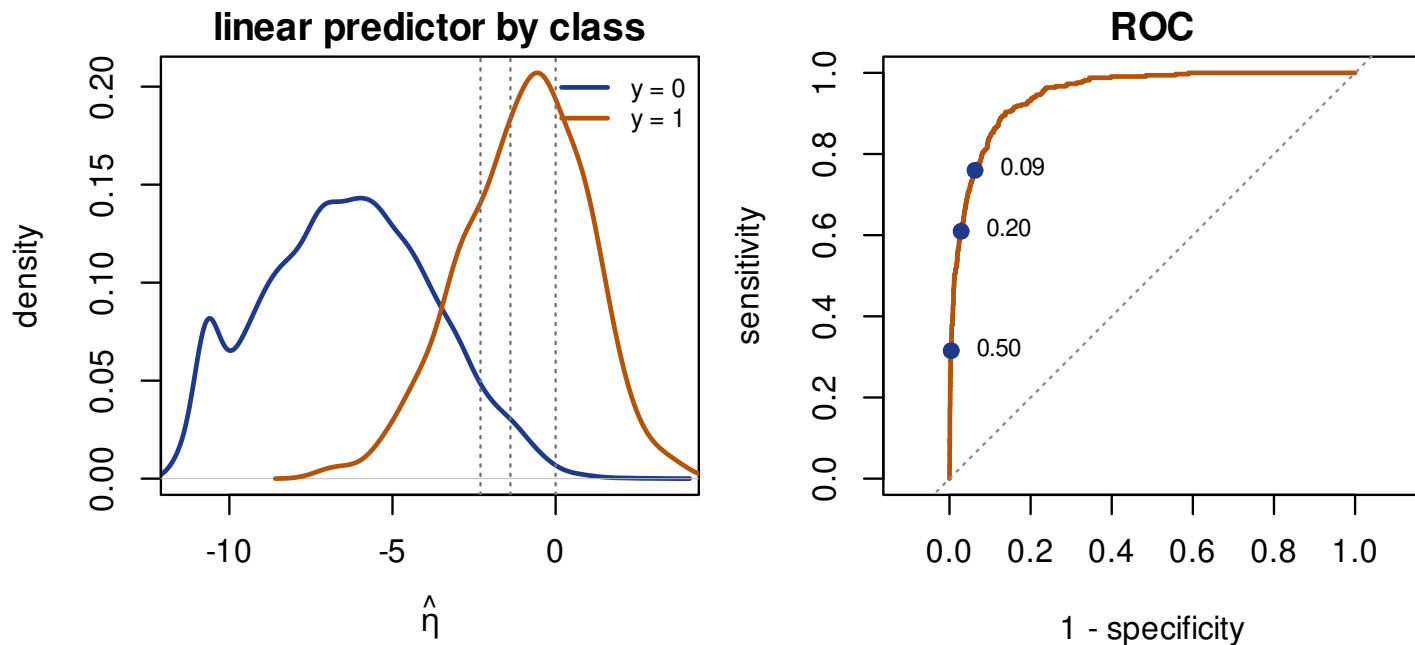


Figure 7.2: Left: the fitted linear predictor by observed class, with the three thresholds of the table marked. Right: the ROC curve (16) with the same three thresholds; the dotted diagonal is the curve of a score carrying no information (Corollary 7.27(ii)).

Theorem 7.23 is a statement about the true conditional probability, so it is checked on data simulated from a known logistic model.

```
1 set.seed(3); N <- 200000
2 xg <- rnorm(N); pt <- plogis(-2 + 1.5*xg); yg <- rbinom(N, 1, pt)
3 grid <- seq(0.005, 0.9, by = 0.005)
4 t(sapply(c(1, 5, 10, 50), function(r) {
5   cs <- sapply(grid, function(t) {
6     yh <- as.numeric(pt > t)
7     r*sum(yh == 0 & yg == 1) + sum(yh == 1 & yg == 0) })
8   c(cost_ratio = r, empirical_argmin = grid[which.min(cs)],
9     theory = 1/(1 + r)) })))
```

```
##      cost_ratio empirical_argmin  theory
## [1,]          1           0.515 0.50000
## [2,]          5           0.165 0.16667
## [3,]         10           0.095 0.09091
## [4,]         50           0.015 0.01961
```

4.6 Nominal responses (§3.1)

```

1 w <- Wage
2 m <- multinom(maritl ~ age + education, data = w, trace = FALSE)
3 B <- coef(m); Xm <- model.matrix(~ age + education, data = w)
4 Vm <- cbind(0, Xm %*% t(B)); Pm <- exp(Vm)/rowSums(exp(Vm))
5 P <- predict(m, type = "probs"); yi <- as.integer(w$maritl)
6 c(rows_sum_to_one = max(abs(rowSums(P) - 1)),
7   manual_vs_predict = max(abs(Pm - P)),
8   manual_loglik = sum(log(Pm[cbind(seq_len(nrow(Pm)), yi)])),
9   multinom_loglik = as.numeric(logLik(m)),
10  free_parameters = length(B),
11  c_minus_1_times_p = (nlevels(w$maritl) - 1)*ncol(Xm))

```

```

## rows_sum_to_one manual_vs_predict manual_loglik
## 2.220e-16 3.331e-16 -2.247e+03
## multinom_loglik free_parameters c_minus_1_times_p
## -2.247e+03 2.400e+01 2.400e+01

```

Corollary 7.32 again holds exactly, and Proposition 7.29 says the choice of baseline is immaterial.

```

1 Y <- model.matrix(~ maritl - 1, data = w)

```

```
2 rbind(observed = colSums(Y), fitted = colSums(P))
```

```
##          maritl1. Never Married maritl2. Married
## observed                648          2074
## fitted                  648          2074
##          maritl3. Widowed maritl4. Divorced
## observed                 19          204
## fitted                  19          204
##          maritl5. Separated
## observed                 55
## fitted                  55
```

```
1 w2 <- w; w2$maritl <- relevel(w$maritl, ref = "3. Widowed")
2 m2 <- multinom(maritl ~ age + education, data = w2, trace = FALSE)
3 c(max_probability_difference =
4   max(abs(predict(m2, type = "probs")[, levels(w$maritl)] - P)),
5   loglik_original = as.numeric(logLik(m)),
6   loglik_rebased = as.numeric(logLik(m2)))
```

```
## max_probability_difference          loglik_original
##                0.00009353          -2247.23798251
```

```
##          loglik_rebased
##          -2247.23800602
```

Theorem 7.33 says the conditional model on two categories is a binary logistic regression with coefficient $\beta_j - \beta_h$, so restricting the sample must reproduce the difference of the full fit.

```
1 sub <- droplevels(subset(w, maritl %in% c("2. Married", "4. Divorced")))
2 gb  <- glm(I(maritl == "4. Divorced") ~ age + education, data = sub,
3         family = binomial)
4 round(rbind(from_the_multinomial = B["4. Divorced", ] - B["2. Married", ],
5         two_category_logistic = coef(gb)), 4)
```

```
##          (Intercept)    age education2. HS Grad
## from_the_multinomial   -3.547 0.0250              0.1988
## two_category_logistic  -3.611 0.0264              0.1999
##          education3. Some College
## from_the_multinomial              0.3289
## two_category_logistic              0.3241
##          education4. College Grad
## from_the_multinomial             -0.0824
## two_category_logistic             -0.0853
```

```
##                                education5. Advanced Degree
## from_the_multinomial          -0.3798
## two_category_logistic         -0.3716
```

4.7 Ordinal responses (§3.2)

```
1 o <- polr(education ~ age + jobclass + health, data = w, Hess = TRUE)
2 Xo <- model.matrix(~ age + jobclass + health, data = w)[, -1, drop = FALSE]
3 al <- unname(o$zeta); bb <- -unname(o$coefficients) # FREES parametrisation
4 cum <- sapply(al, function(a) plogis(a + drop(Xo %*% bb)))
5 pm <- cbind(cum, 1) - cbind(0, cum); yo <- as.integer(w$education)
6 c(manual_loglik = sum(log(pm[cbind(seq_len(nrow(pm)), yo]))),
7   polr_loglik = as.numeric(logLik(o)),
8   probabilities_vs_predict = max(abs(pm - predict(o, type = "probs"))),
9   cut_points_increasing = all(diff(al) > 0))
```

```
##                manual_loglik                polr_loglik
##                -4392                -4392
## probabilities_vs_predict    cut_points_increasing
##                0                1
```

```
1 round(rbind(beta_of_equation_polr = bb,
```

```
2 gamma_reported_by_polr = unname(o$coefficients)), 5)
```

```
##                [,1]    [,2]    [,3]
## beta_of_equation_polr -0.01067 -1.085 -0.7026
## gamma_reported_by_polr 0.01067  1.085  0.7026
```

Corollary 7.38: the four dichotomisations $\mathbb{1}\{y \leq j\}$ are logistic regressions sharing one slope vector under the model.

```
1 sep <- sapply(1:4, function(j)
2   coef(glm(I(as.integer(education) <= j) ~ age + jobclass + health,
3     data = w, family = binomial))[-1])
4 colnames(sep) <- paste0("y<=", 1:4)
5 round(cbind(sep, proportional_odds = bb), 4)
```

```
##                y<=1    y<=2    y<=3    y<=4
## age            -0.0049 -0.0034 -0.0161 -0.0244
## jobclass2. Information -0.8660 -1.0407 -1.1083 -1.3327
## health2. >=Very Good  -0.5425 -0.6553 -0.8014 -0.8570
##                proportional_odds
## age                        -0.0107
## jobclass2. Information      -1.0848
```

```
## health2. >=Very Good          -0.7026
```

The slopes drift steadily across the cut points instead of agreeing, so the proportional odds restriction is doubtful here. The two families are not nested (§3.2), so they are compared by an information criterion.

```
1 mo <- multinom(education ~ age + jobclass + health, data = w, trace = FALSE)
2 c(polr_loglik = as.numeric(logLik(o)),
3   polr_parameters = length(coef(o)) + length(o$zeta), polr_AIC = AIC(o),
4   multinom_loglik = as.numeric(logLik(mo)),
5   multinom_parameters = length(coef(mo)), multinom_AIC = AIC(mo))
```

##	polr_loglik	polr_parameters	polr_AIC
##	-4392	7	8797
##	multinom_loglik	multinom_parameters	multinom_AIC
##	-4375	16	8782

5 Pitfalls

5.1 Reading a separated fit as a strong effect

Theorem 7.14 says that when a linear combination of the covariates separates the two outcomes the maximum likelihood estimate does not exist. Software returns a number anyway.

```
1 xs <- c(1, 2, 3, 4, 5, 6); ys <- c(0, 0, 0, 1, 1, 1)
2 gs <- suppressWarnings(glm(ys ~ xs, family = binomial))
3 c(slope = unname(coef(gs)[2]), se = summary(gs)$coefficients[2, 2],
4   z = summary(gs)$coefficients[2, 3], deviance = deviance(gs),
5   iterations = gs$iter, min_fitted = min(fitted(gs)),
6   max_fitted = max(fitted(gs)))
```

```
##      slope      se      z  deviance iterations
## 4.723e+01 1.153e+05 4.098e-04 2.215e-10 2.500e+01
## min_fitted max_fitted
## 2.220e-16 1.000e+00
```

```
1 loglik <- function(t) { z <- t*(xs - 3.5)
2   sum(ys*plogis(z, log.p = TRUE) + (1 - ys)*plogis(-z, log.p = TRUE)) }
3 rbind(slope = c(1, 5, 20, 100, 1000),
4       loglik = sapply(c(1, 5, 20, 100, 1000), loglik))
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]
## slope    1.000    5.0000 20.0000000 1.000e+02 1.000e+03
## loglik -1.509 -0.1589 -0.0000908 -3.857e-22 -1.425e-217
```

Diagnostic. A very large coefficient with a far larger standard error, a z statistic near zero, a residual deviance near zero, fitted probabilities at 0 and 1, and the iteration limit approached. **Fix.** Report the separation itself, which is the finding; or penalise the likelihood, for instance by the ridge penalty of Unit 6 applied to ℓ , which restores a maximum because the penalised objective is coercive.

5.2 Comparing coefficients across models

In the linear model, dropping a covariate that is independent of the others leaves their coefficients alone (Unit 3). Logistic coefficients move anyway.

Proposition 7.39 (Non-collapsibility). Let x_2 take the values ± 1 with probability $\frac{1}{2}$ each, independently of x_1 , and let $\mathbf{P}\{y = 1 \mid x_1, x_2\} = \Lambda(\beta_1 x_1 + \beta_2 x_2)$. Then $\mathbf{P}\{y = 1 \mid x_1\} = \Psi(\beta_1 x_1)$ with $\Psi(u) = \frac{1}{2}\{\Lambda(u + \beta_2) + \Lambda(u - \beta_2)\}$, and

$$\left. \frac{\partial}{\partial x_1} \text{logit } \mathbf{P}\{y = 1 \mid x_1\} \right|_{x_1=0} = 4\Lambda'(\beta_2) \beta_1 = \frac{4e^{\beta_2}}{(1 + e^{\beta_2})^2} \beta_1, \quad (26)$$

whose modulus is strictly less than $|\beta_1|$ unless $\beta_2 = 0$.

Proof. The tower property gives $\mathbf{P}\{y = 1 \mid x_1\} = \mathbf{E}[\Lambda(\beta_1 x_1 + \beta_2 x_2) \mid x_1] = \Psi(\beta_1 x_1)$, the independence of x_1 and x_2 being what removes x_1 from the averaging. Write $q(x_1) = \Psi(\beta_1 x_1)$. Then $\partial \logit q / \partial x_1 = q' / \{q(1 - q)\}$ with $q' = \beta_1 \Psi'(\beta_1 x_1)$. At $x_1 = 0$, $\Lambda(\beta_2) + \Lambda(-\beta_2) = 1$ by Proposition 7.5, so $q = \frac{1}{2}$ and $q(1 - q) = \frac{1}{4}$. Also Λ' is even, since $\Lambda'(-z) = \Lambda(-z)\{1 - \Lambda(-z)\} = \{1 - \Lambda(z)\}\Lambda(z)$, so $\Psi'(0) = \frac{1}{2}\{\Lambda'(\beta_2) + \Lambda'(-\beta_2)\} = \Lambda'(\beta_2)$, giving (26). Finally $4\Lambda'(t) = 4e^t/(1 + e^t)^2$ equals 1 at $t = 0$ and is strictly smaller elsewhere, because $(1 + e^t)^2 - 4e^t = (1 - e^t)^2 > 0$ for $t \neq 0$. $\square$

```

1 b1 <- 1
2 t(sapply(c(0.5, 1, 2, 3), function(b2) {
3   Psi <- function(u) (plogis(u + b2) + plogis(u - b2))/2
4   h <- 1e-5
5   c(beta2 = b2,
6     numeric_slope = (qlogis(Psi(b1*h)) - qlogis(Psi(-b1*h)))/(2*h),
7     formula = 4*plogis(b2)*(1 - plogis(b2))*b1) })))

```

```
##      beta2 numeric_slope formula
```

```
## [1,] 0.5 0.9400 0.9400
## [2,] 1.0 0.7864 0.7864
## [3,] 2.0 0.4200 0.4200
## [4,] 3.0 0.1807 0.1807
```

```
1 set.seed(9); N <- 400000; x1 <- rnorm(N)
2 t(sapply(c(0, 1, 2, 3), function(b2) {
3   x2 <- sample(c(-1, 1), N, TRUE)
4   yy <- rbinom(N, 1, plogis(b1*x1 + b2*x2))
5   c(beta2 = b2,
6     logistic_with_x2 = coef(glm(yy ~ x1 + x2, family = binomial))[2],
7     logistic_without = coef(glm(yy ~ x1, family = binomial))[2],
8     linear_with_x2 = coef(lm(yy ~ x1 + x2))[2],
9     linear_without = coef(lm(yy ~ x1))[2]) })))
```

```
##      beta2 logistic_with_x2.x1 logistic_without.x1
## [1,] 0 1.0017 1.0017
## [2,] 1 0.9964 0.8140
## [3,] 2 0.9964 0.4918
## [4,] 3 1.0090 0.2424
##      linear_with_x2.x1 linear_without.x1
```

## [1,]	0.20695	0.20695
## [2,]	0.17744	0.17766
## [3,]	0.11539	0.11631
## [4,]	0.06004	0.05974

Diagnostic. A logistic coefficient that changes when an unrelated covariate is added or removed, with no correlation between the two covariates to blame. **Fix.** Do not read the change as confounding: it is the odds ratio failing to be collapsible. Compare models on the probability scale, through fitted values or average marginal effects, or fix the covariate set before interpreting anything.

5.3 Accuracy at threshold one half

```

1 c(accuracy_at_half = mean((s > 0.5) == (y == 1)),
2   accuracy_predicting_zero = mean(y == 0),
3   sensitivity_at_half = sum(s > 0.5 & y == 1)/sum(y == 1),
4   AUC = rank_auc(s))

```

##	accuracy_at_half	accuracy_predicting_zero
##	0.9732	0.9667
##	sensitivity_at_half	AUC
##	0.3153	0.9496

Diagnostic. High accuracy on an imbalanced response, with sensitivity far below it, and a threshold that was never chosen but inherited. **Fix.** Choose the threshold from the costs by Theorem 7.23, and report sensitivity and specificity, or the whole ROC curve, alongside any single number. Accuracy is the average of specificity and sensitivity weighted by the class proportions, here 0.967 and 0.033, so a rule can be accurate while finding almost none of the positives.

5.4 Assuming proportional odds

The restriction of Theorem 7.37 is testable and is frequently false. Fitting `polr` without checking it produces one slope where the data ask for several, and the single slope is an average that describes no cut point.

```
1 round(cbind(sep, proportional_odds = bb), 4)
```

```
##                y<=1    y<=2    y<=3    y<=4
## age            -0.0049 -0.0034 -0.0161 -0.0244
## jobclass2. Information -0.8660 -1.0407 -1.1083 -1.3327
## health2. >=Very Good  -0.5425 -0.6553 -0.8014 -0.8570
##                proportional_odds
## age                        -0.0107
## jobclass2. Information     -1.0848
```

```
## health2. >=Very Good          -0.7026
```

```
1 c(jobclass_slope_max_over_min = max(abs(sep[2, ])/min(abs(sep[2, ])),
2   polr_AIC = AIC(o), multinom_AIC = AIC(mo))
```

```
## jobclass_slope_max_over_min      polr_AIC
##                               1.539      8797.225
##               multinom_AIC
##               8782.489
```

Diagnostic. Slopes from the $c - 1$ dichotomisations that drift monotonically across j rather than scattering about a common value. **Fix.** Report the separate fits, or use the generalised logit, which imposes no ordering and no common slope, at the cost of $(c - 2)k$ extra parameters and of discarding the ordering that the response actually has.

6 Exercises

Exercise 7.1. Give three reasons not to model a binary response by least squares on the covariates, and say which of them a weighted least squares fit repairs.

Solution. Proposition 7.2: (i) the fitted mean is a probability but $\mathbf{x}'\boldsymbol{\beta}$ is unbounded, so the model cannot hold over the whole covariate space and the fitted values need not lie in $[0, 1]$; (ii) the errors are heteroscedastic by construction, with variance $\pi_i(1 - \pi_i)$; (iii) where a fitted value leaves $[0, 1]$ the variance the model itself implies is negative.

Weighted least squares with weights $1/\{\pi_i(1 - \pi_i)\}$, or the robust standard errors of Unit 4, repair (ii) alone. Defects (i) and (iii) are properties of the mean specification and survive any reweighting.



Exercise 7.2. A logistic regression of default on a 0/1 indicator of being a student gives coefficient 0.40. State the odds ratio, and compute the change in the probability of default that this implies for a customer whose probability is 0.02 and for one whose probability is 0.50.

Solution. By Theorem 7.8(i) the odds ratio is $e^{0.40} = 1.492$: the odds of default are about 49% higher for students.

For the probabilities, work on the odds scale. At $\pi = 0.02$ the odds are $0.02/0.98 = 0.02041$; multiplying by 1.492 gives 0.03046, hence $\pi' = 0.03046/1.03046 = 0.02956$, a rise of 0.0096. At $\pi = 0.50$ the odds are 1; multiplying gives 1.492 and $\pi' = 1.492/2.492 = 0.5987$, a rise of 0.0987, ten times as large. The derivative approximation of Theorem 7.8(iii) predicts $0.40 \times 0.02 \times 0.98 = 0.0078$ and $0.40 \times 0.25 = 0.10$ respectively, close in the second case and less so in the first, where the change is not small on the odds scale. ■

Exercise 7.3. Prove that the logistic log-likelihood is concave, and that the probit log-likelihood is too. Where does the probit argument need a bound on the normal tail?

Solution. Both are Theorem 7.13. For the logistic case $(\ln \Lambda)'' = -\Lambda(1 - \Lambda) < 0$ directly. For the probit case the requirement reduces to $\eta\Phi(\eta) + \phi(\eta) > 0$ for every η , which is immediate for $\eta \geq 0$ and, for $\eta = -t < 0$, is the Mills bound $1 - \Phi(t) < \phi(t)/t$. That bound is proved by writing $1 - \Phi(t) = \int_t^\infty \phi$ and using $s/t > 1$ on (t, ∞) to compare with $\int_t^\infty (s/t)\phi(s) ds = \phi(t)/t$. The tail bound is exactly what rules out the log-likelihood turning convex far out in the tail, where Φ

is exponentially small and its logarithm behaves like $-\eta^2/2$. ■

Exercise 7.4. Six observations have $x = 1, \dots, 6$ and $y = 0, 0, 0, 1, 1, 1$. Show that no maximum likelihood estimate exists, and say what a package reports.

Solution. Take $\mathbf{c} = (-3.5, 1)'$, so that $\mathbf{x}_i' \mathbf{c} = x_i - 3.5$, which is negative for $x_i \leq 3$ and positive for $x_i \geq 4$: the data are separated in the sense of (8). By Theorem 7.14 the log-likelihood along $t\mathbf{c}$ increases strictly to 0, and since $\ell < 0$ everywhere the supremum 0 is not attained.

A package stops at its iteration or tolerance limit and reports a large coefficient with a much larger standard error, a residual deviance numerically zero, and fitted probabilities of exactly 0 and 1; §5.1 prints all four. ■

Exercise 7.5. Show that the Newton step for the logistic model is a weighted least squares fit, identify the weights and the working response, and say why Newton–Raphson and Fisher scoring coincide here.

Solution. Theorem 7.15: the weights are $w_i = \pi_i(1 - \pi_i)$ and the working response is $z_i = \eta_i + (y_i - \pi_i)/w_i$, and the update is $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{z}$. The two algorithms coincide because for

the logistic model the Hessian $-\nabla^2\ell = \sum_i \pi_i(1 - \pi_i)\mathbf{x}_i\mathbf{x}_i'$ contains no y_i (Theorem 7.11), so the observed information already equals the expected information and there is nothing for Fisher scoring to replace. ■

Exercise 7.6. A logistic fit reports $\hat{\beta} = 0.65$ with standard error 0.24; dropping the variable raises the deviance by 7.10. Compute the Wald statistic and the likelihood ratio statistic, and explain why they differ.

Solution. $W = (0.65/0.24)^2 = 7.335$ and $LR = 7.10$, the latter because a deviance difference between nested models **is** the likelihood ratio statistic (Proposition 7.20). Both are referred to χ_1^2 , giving $p = 0.0068$ and $p = 0.0077$.

They differ because they are different functions of the data that happen to share a limiting distribution (Corollary 7.18). The Wald statistic uses a quadratic approximation to ℓ centred at $\hat{\beta}$, and the likelihood ratio uses the actual heights of the two maximised log-likelihoods; the approximation is exact only when ℓ is exactly quadratic, which it is not. The gap grows with $|\hat{\beta}|$, and in the separated case of Theorem 7.14 the Wald statistic collapses to nearly zero while the likelihood ratio does not. ■

Exercise 7.7. For $n = 500$ ungrouped binary observations and a model with $p = 6$ parameters, the residual deviance is 412. Is the model adequate?

Solution. The question cannot be answered from that number. By Proposition 7.20 the deviance of an ungrouped binary fit is $-2\ell(\hat{\beta})$, a function of the fitted probabilities alone; the saturated model it is measured against has one parameter per observation, so its parameter count grows with n and the asymptotics behind the χ^2 approximation do not apply. Comparing 412 with $n - p = 494$ is meaningless.

What is available: differences of deviances between nested models, which are likelihood ratio statistics, and out-of-sample assessment. ■

Exercise 7.8. A missed default costs 20 times a false alarm. What threshold should the fitted probability be compared against, and what happens to sensitivity and specificity as the cost ratio grows?

Solution. By Theorem 7.23, $t^* = c_{10}/(c_{10} + c_{01}) = 1/21 = 0.0476$.

As c_{01}/c_{10} grows the threshold falls towards 0, so more cases are predicted positive: TP and FP

both increase, sensitivity increases towards 1 and specificity falls towards 0. In the limit every case is predicted positive, which is optimal when a miss is infinitely expensive. The ROC curve is the locus of the attainable pairs, and moving the threshold moves along it. ■

Exercise 7.9. Show that the area under the ROC curve equals $\mathbf{P}\{S^+ > S^-\}$, and deduce that dividing every fitted probability by 10 leaves the area unchanged. Is that a virtue?

Solution. Theorem 7.26: parametrise the curve by $u = 1 - F^-(t)$, so that $\text{AUC} = \int \mathbf{P}\{S^+ > t\} dF^-(t) = \mathbf{P}\{S^+ > S^-\}$ by independence and conditioning on S^- . Since $s \mapsto s/10$ is strictly increasing, Corollary 7.27(i) applies and the area is unchanged.

It is a virtue for comparing rankings and a defect for anything else. A model whose fitted probabilities are all ten times too small ranks cases perfectly well and is useless for the decision of Theorem 7.23, which compares a probability with a threshold. Ranking and calibration are separate properties and AUC measures only the first. ■

Exercise 7.10. In a generalised logit with four categories, explain why adding a constant vector to every β_j changes nothing, and state what the model implies about the odds between two categories

when a third is removed from the choice set.

Solution. Proposition 7.29: multiplying numerator and denominator of (19) by $e^{-\mathbf{x}'_i \mathbf{a}}$ shows the probabilities depend on the β_j only through their differences, so the model has $(c - 1)p$ free parameters and the normalisation $\beta_c = \mathbf{0}$ is free.

By Theorem 7.33 the odds π_{ij}/π_{ih} are $\exp\{\mathbf{x}'_i(\beta_j - \beta_h)\}$, which mentions no other category: removing one leaves the odds between the survivors untouched. That is independence of irrelevant alternatives, and it is implausible whenever two categories are close substitutes, since removing one of a pair of substitutes should send its share mostly to the other. ■

Exercise 7.11. State the proportional odds model, prove the interpretation of β_l , and describe a check of the restriction that needs only binary logistic regressions.

Solution. The model is (24), $\text{logit } \mathbf{P}\{y_i \leq j\} = \alpha_j + \mathbf{x}'_i \boldsymbol{\beta}$ with $\alpha_1 < \dots < \alpha_{c-1}$, the ordering being what keeps every category probability positive (Proposition 7.35).

By Theorem 7.37, a unit increase in x_{il} multiplies the cumulative odds $\text{odds}\{y_i \leq j\}$ by e^{β_l} , and the factor is the same at every cut point j because α_j cancels in the difference.

The check is Corollary 7.38: under the model, $\mathbf{1}\{y_i \leq j\}$ is a logistic regression on $\mathbf{x}_i$ with slope β for every j , so fitting the $c - 1$ dichotomisations separately gives $c - 1$ estimates of one vector. Systematic drift across j contradicts the restriction; §5.4 shows such a drift. ■

Exercise 7.12. In R: (a) reproduce a `glm` logistic fit by iteratively reweighted least squares; (b) compute the area under the ROC curve three ways; (c) show that a logistic coefficient changes when an independent covariate is dropped, while a linear one does not.

Solution. (a) is §4.2, (b) §4.5 and (c) §5.2. In (a) the point is that no optimiser is needed: each step is one weighted least squares solve, and the concavity of Theorem 7.13 guarantees there is a single stationary point to converge to. In (c) note that the two covariates are independent by construction, so the omitted-variable bias formula of Unit 3 predicts no change, and the linear fit shows none; the logistic change is Proposition 7.39. ■